Automated Artwork Metadata Generation using Multimodal LLM

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Abstract

In this work, we propose an AI-based solution for automated artwork metadata generation, leveraging a Multimodal Large Language Model (MLLM). The solution is built upon a dataset derived from Wikidata, an open-source knowledge base, and utilizes Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically Low-Rank Adaptation (LoRA), to adapt a pre-trained model to generate rich, multi-label metadata across multiple categories. This approach addresses the scalability and computational challenges associated with large multimodal models in the cultural heritage domain. Experimental results demonstrate the model's effectiveness in producing accurate and contextually relevant metadata, showing significant improvements over existing approaches. You can find our project at:

https://github.com/adsp-polito/2025-P4-Artwork-Metadata-Generation

1. Introduction

The large-scale digitization of cultural heritage collections has become a central activity for museums, archives, and galleries. By converting physical artworks into digital resources, institutions aim to improve preservation, enable research, and broaden public access. However, this transformation introduces a major challenge: the creation of accurate and consistent descriptive metadata for large numbers of digital artworks.

Metadata annotation is a time-consuming and costly process for curators and does not scale to collections containing thousands or millions of objects. At the same time, automated solutions face significant obstacles, including limited availability of high-quality labeled data and strong heterogeneity in metadata schemas across institutions. These factors make it difficult to design scalable and transferable systems for artwork description.

In this work, we address automated metadata generation by applying fine-tuned multimodal large language models (MLLMs) to artwork images. The task, shown in Figure 1, is formulated as a generative, multi-task, multi-label problem, where a single model produces multiple types of metadata such as authors, genres, subjects, and materials, in natural language. Each field may contain multiple valid labels, reflecting the inherent complexity of the artworks.

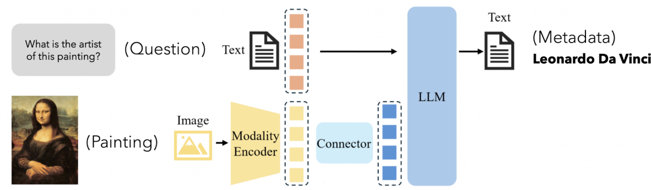

Figure 1. Multimodal Large Language Model generating metadata

To ensure practicality in large models and real institutional collections, our approach leverages parameter-efficient fine-tuning (PEFT) techniques [1], specifically Low-Rank Adaptation (LoRA) [3]. The dataset used in this work is constructed from data sourced from Wikidata [15].

This work focuses on artwork metadata extraction by introducing a unified generative multi-task framework in which a single multimodal language model simultaneously predicts heterogeneous metadata fields across multiple taxonomic categories, eliminating the need for task-specific architectures. To address the computational challenges inherent in large-scale metadata generation, we propose a parameter-efficient adaptation strategy grounded in Low-Rank Adaptation (LoRA), which substantially reduces training costs and memory consumption while maintaining scalability for deployment in resource-constrained environments. We further develop and release a rigorously curated benchmark dataset, enabling comprehensive evaluation of metadata generation performance across four critical dimensions: artist attribution, genre classification, material composition, and subject matter identification.

2. Related Work

In the context of artificial intelligence applied to cultural heritage, several computational approaches address artwork analysis and metadata extraction. Early investigations employ convolutional neural networks (CNNs) with shared feature extractors and task-specific classification heads for narrow tasks such as style classification, artist attribution, and genre prediction [2, 14].

Despite their effectiveness on controlled datasets, these discriminative architectures demonstrate limited generalization across heterogeneous digitized collections, primarily due to domain shift introduced by varying acquisition and imaging conditions. Following the emergence of multimodal large language models (MLLMs), research shifts toward unified visual-linguistic frameworks that enable richer semantic understanding. Radford et al. [12] introduce CLIP, aligning vision and language representations through contrastive learning and establishing a foundation for cross-modal reasoning.

Liu et al. [10] propose LLaVA, integrating a CLIP-derived vision encoder with an autoregressive language model to enable flexible instruction-following and generation. Recent domain-specific applications demonstrate the potential of these architectures for cultural heritage: Lee et al. [8] adapt MLLMs with LLaVA-Docent for art interpretation and educational engagement, while GalleryGPT [6] shows that large multimodal models can generate coherent analyses of paintings by jointly reasoning over visual and textual representations.

Similarly, ArtGPT-4 [7] demonstrates that adapter-based architectures significantly enhance artistic understanding in vision-language models.

Despite these advances, existing MLLM-based methods primarily focus on qualitative description generation rather than structured metadata extraction, leaving a gap in systematic multi-category prediction frameworks for cultural heritage documentation.

3. Methodology

We formalize the artwork metadata extraction task as a conditional multimodal generation problem. Let $\mathcal{D} = (x_i, M_i)i = 1^N$ denote a dataset of N artwork instances, where each $x_i \in \mathbb{R}^{H\times W\times 3}$ represents a high-resolution digitized artwork image, and $M_i = m_i^{(k)}k = 1^K$ denotes the corresponding structured metadata across K categories (authors, genres, materials, subjects).

Our objective is to learn a multimodal function $f_\theta : \mathcal{X} \times \mathcal{C} \rightarrow \mathcal{Y}$ that maps an artwork image $x \in \mathcal{X}$ and a category textual prompt $c \in \mathcal{C}$ to a target metadata sequence $y \in \mathcal{Y}$. This is formalized as:

$$y^* = MLLM_\theta(x, c), \tag{1}$$

where $y^* = (y_1^*, y_2^*, \ldots, y_T^*)$ is a sequence of predicted tokens representing the metadata for category c, θ denotes the trainable parameters of the model, and $MLLM_\theta(x, c)$ represents a multimodal large language model conditioned on an image input x and a textual prompt c.

The input image x is processed by a pretrained vision encoder $E_v : \mathbb{R}^{H\times W\times 3} \rightarrow \mathbb{R}^{n\times d_v}$, which produces a sequence of n visual tokens:

$$\mathbf{v} = E_v(x) = [v_1, v_2, \ldots, v_n] \in \mathbb{R}^{n\times d_v}, \tag{2}$$

where each $v_i \in \mathbb{R}^{d_v}$ captures local visual features essential for identifying stylistic and compositional characteristics of the artwork, and d_v denotes the dimensionality of the visual token embeddings.

Category prompts c are encoded into textual embeddings through a learned embedding function $E_t : \mathcal{C} \rightarrow \mathbb{R}^{m\times d_t}$:

$$\mathbf{t} = E_t(c) = [t_1, t_2, \ldots, t_m] \in \mathbb{R}^{m\times d_t}, \tag{3}$$

where m denotes the prompt sequence length and d_t is the text embedding dimension.

The multimodal representation $\mathbf{z} \in \mathbb{R}^{s\times d}$ is formed by aligning the visual and textual tokens, where s is the sequence length after alignment and d is the dimensionality of the resulting multimodal embedding. This representation is then processed by LLM_θ, an autoregressive language model decoder, which generates the output metadata sequence token by token:

$$y_t = LLM_\theta(y_{<t}, \mathbf{z}), \tag{4}$$

where $y_{<t}$ denotes the previously generated tokens up to timestep $t-1$.

To adapt the pretrained MLLM to the artwork domain while minimizing computational costs, we employ Low-Rank Adaptation (LoRA). For each weight matrix $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$ and $r \ll \min(d_{\text{in}}, d_{\text{out}})$ in the attention and feed-forward layers, LoRA introduces trainable low-rank matrices $A \in \mathbb{R}^{d_{\text{in}}\times r}$ and $B \in \mathbb{R}^{r\times d_{\text{out}}}$:

$$W' = W + \Delta W = W + AB, \tag{5}$$

where W remains frozen during training and only A, B are optimized.

The model is trained to minimize the negative log-likelihood over all metadata categories and instances:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N}\sum_{k=1}^{K}\sum_{t=1}^{T_{ik}} \log P_\theta(y_{i,k,t} \mid y_{i,k,<t}, \mathbf{z}_{i,k}), \tag{6}$$

where T_{ik} is the sequence length for category k in instance i, and $\mathbf{z}_{i,k}$ is the multimodal representation for that instance-category pair. This optimization enables the model to learn shared visual-semantic representations across multiple metadata categories simultaneously.

4. Dataset

The dataset we construct draws its information from Wikidata [15], an extensive, open-source knowledge repository that provides structured data across diverse domains,

	Unique elements		Total occurrences		Avg elements/row	
	TRAIN	VAL	TRAIN	VAL	TRAIN	VAL
Authors	3,282	1,040	7,372	1,773	1.02	1.01
Genres	87	41	7,601	1,882	1.05	1.07
Materials	110	47	14,047	3,494	1.94	1.99
Subjects	6,217	2,442	20,497	6,446	2.83	3.67

Table 1. Statistics of the dataset across different categories for Train and Validation splits.

including the fine arts. This corpus consists of artworks paired with both high-resolution image data and corresponding textual metadata. The dataset comprises a total of 9,008 artworks, partitioned into 7,252 samples for training and 1,756 samples for validation. This split is designed to provide a robust framework for model fitting, while ensuring sufficient data to evaluate the model's ability to generalize to unseen instances.

Each entry is associated with a broad range of labels in four primary metadata fields: authors, genres, materials, and subjects.

Table 1 provides a comprehensive summary of the label space and the multi-label distribution across the four metadata fields. Statistical analysis of the combined training and validation sets reveals a highly imbalanced distribution. The dataset is imbalanced, as the genres and materials columns contain relatively few labels, while the authors and subjects columns feature a significantly larger number of labels.

This represents a significant departure from standard single-label classification tasks, as it requires a model capable of predicting multiple concurrent attributes for a single input.

5. Experiments and Results

5.1. Experimental Setup

The experimental phase evaluates the effectiveness of different architectural configurations and fine-tuning strategies for generating structured metadata across four categories: authors, genres, materials, and subjects.

To identify the most suitable backbone architecture for artwork metadata extraction, we conduct a preliminary zero-shot comparative evaluation between Qwen2-VL-2B-Instruct [5] and BLIP-2 (Flan-T5-XL) [4]. Both models are evaluated on the validation set without fine-tuning, using a distinct prompt tailored to each metadata category.

The metadata generation task is formulated as an open-vocabulary classification problem. Unlike traditional closed-set classification, where the model is limited to a predefined list of categories, an open-vocabulary approach allows the system to recognize and generate labels that were not necessarily seen during training. Consequently, conventional closed-set metrics such as Precision, Recall, and F1-score are no longer sufficient and are inherently inadequate for this context. Thus, we evaluate the performance considering the following metrics:

- **Exact Match (EM):** Quantifies the percentage of generated metadata fields identical to ground-truth reference labels from Wikidata. For the Authors category, we extend EM by cross-referencing Wikidata aliases to avoid penalizing valid naming alternatives (e.g., recognizing *"Canal"* as a valid alias for *"Canaletto"*).

- **Semantic Similarity:** To account for semantically equivalent but syntactically distinct outputs, we compute cosine similarity between predicted and reference labels using MPNet [13] embeddings. This provides a nuanced understanding of conceptually accurate predictions that do not meet exact match criteria.

- **BLEU and ROUGE:** These n-gram overlap metrics evaluate the linguistic fluency and structural completeness of generated metadata [9, 11].

Following the zero-shot evaluation of the generative pipeline, we selected the best-performing model and further optimized it using Parameter-Efficient Fine-Tuning (PEFT). We employ Low-Rank Adaptation (LoRA) [3] to inject trainable low-rank matrices into the transformer layers, significantly reducing computational overhead and memory footprint while maintaining performance. Table 3 summarizes the LoRA hyperparameter configurations investigated across single-model strategies.

Subsequently, we evaluated three distinct operational strategies, corresponding to different prompt formulations, to analyze the trade-offs between model complexity, inference efficiency, and prediction accuracy:

- **Single-Model, Single Prompt:** A monolithic configuration where a single model instance generates all four metadata fields simultaneously within one inference pass. The model receives a comprehensive prompt requesting all categories at once.

- **Single-Model, 1 IMG-4 PROMPT:** A single model processes the same artwork image iteratively across

	Exact Match		Similarity Score		BLEU		ROUGE-1	
	QWEN	BLIP	QWEN	BLIP	QWEN	BLIP	QWEN	BLIP
Authors	**0.150**	0.001	**0.800**	0.684	X	X	**0.154**	0.003
Subjects	**0.014**	0.000	0.717	**0.722**	**0.030**	0.014	**0.193**	0.099
Materials	**0.218**	0.000	**0.886**	0.838	**0.210**	0.105	**0.609**	0.478
Genres	0.064	**0.353**	**0.586**	0.549	**0.079**	0.077	0.324	**0.485**

Table 2. Comparison of QWEN and BLIP across different evaluation metrics.

four inference passes, with each pass utilizing a category-specific prompt to isolate task-focused reasoning. This approach enables specialized prompting strategies per metadata category.

- **Multi-Model :** Four independent, category-specific models, each fine-tuned exclusively for one metadata category. This configuration represents the performance upper bound for task specialization, as each model is optimized for a single category.

Configuration	Rank (r)	Dropout (D)
A	8	0.1
B	16	0.05

Table 3. LoRA hyperparameter configurations.

The model is optimized using cross-entropy loss for autoregressive next-token prediction:

$$\mathcal{L} = -\sum_{t=1}^{T} \log P(y_t \mid y_{<t}, x), \tag{7}$$

where x represents the input artwork image and $y = (y_1, \ldots, y_T)$ is the generated metadata sequence.

All experiments are conducted on NVIDIA A100 GPUs and training employs mixed-precision (FP16) to accelerate convergence and reduce memory consumption.

5.2. Results and Discussion

We now present the quantitative performance of our approach across different architectural strategies and hyperparameter settings.

5.2.1 Zero-shot evaluation

This section evaluates the zero-shot capabilities of the considered multimodal models.

Table 2 reports the zero-shot performance across all evaluation metrics, showing that Qwen2-VL achieves superior results in all metrics. Based on these results, Qwen2-VL is selected as the primary backbone for all subsequent fine-tuning experiments.

5.2.2 Fine-Tuning experiments

This section presents the fine-tuning experiments conducted to optimize the multimodal model.

The experimental evaluation is conducted in two sequential phases:

a) we identify the optimal LoRA hyperparameter configuration for both single-model strategies by comparing the two configurations keeping a constant learning rate, as presented in Table 3 .

b) building upon the optimal LoRA configuration, we further refine the training dynamics by integrating a Cosine Learning Rate Scheduler that gradually reduces the learning rate following a cosine annealing schedule, allowing for finer weight updates in the later stages of training.

Table 4 presents the results obtained training the LLM in our pipeline with different ranks and dropout rate, as shown in Table 3. This suggests that the higher rank provides the necessary capacity to capture the complex, multi-label dependencies of artwork metadata without overfitting.

Strategy	Config.	Authors	Genres	Materials	Subjects
1 IMG–4 PROMPT	A	17.71	75.80	68.74	**7.92**
	B	**18.45**	**76.37**	**69.19**	7.74
SINGLE PROMPT	A	17.26	75.34	66.57	6.66
	B	**17.06**	**75.57**	**66.86**	**7.00**

Table 4. Baseline comparison: Impact of Rank (R) and Dropout (D). Bold indicates the best configuration within each strategy. (accuracy in %).

We then apply the learning rate scheduler to the best-performing configuration for both prompt strategies. As evidenced in Table 5, the scheduler consistently improves performance for both approaches across different categories. Specifically, for the 1 IMG–4 PROMPT strategy, we observe notable gains in Materials (+0.40%) and Subjects (+0.40%), whereas the SINGLE PROMPT strategy benefits primarily in the Authors (+0.37%) and Genres (+0.40%) categories. These findings demonstrate the scheduler's effectiveness in refining the optimization trajectory for specific metadata fields.

Focusing on the 1 IMG–4 PROMPT strategy, although the performance trade-off appears balanced—with the scheduler-enabled and baseline configurations each outperforming the other in two categories—we prioritize the configuration equipped with the Cosine Scheduler. This decision is strategically driven by the performance on the Subjects category, as highlighted in the dataset analysis, it represents the most computationally challenging task due to its extreme label sparsity, high cardinality, and semantic complexity.

Strategy	Scheduler	Authors	Genres	Materials	Subjects
1 IMG–4 PROMPT	✗	**18.45**	**76.37**	69.19	7.74
	✓	18.28	75.68	**69.59**	**8.14**
SINGLE PROMPT	✗	17.06	75.57	66.86	**7.00**
	✓	**17.43**	**75.97**	**67.43**	6.66

Table 5. Impact of Cosine Learning Rate Scheduler on Configuration B ($R = 16, D = 0.05$). Bold indicates improvements. (accuracy in %).

The empirical results demonstrate that the 1 IMG–4 PROMPT approach surpasses the Single Prompt strategy in three out of four categories, showing gains in Authors (+0.85%), Materials (+2.16%), and Subjects (+1.48%). The Single-Prompt strategy exhibits a marginal advantage only in the Genres category (+0.29%), confirming the overall effectiveness of the multi-prompt decomposition for complex metadata extraction.

To validate the efficiency of our best model, we first benchmark it against the Multi-Model approach, which serves as our theoretical performance upper bound.

As shown in Table 6, our 1 IMG–4 PROMPT configuration remains remarkably competitive against the specialized upper bound. The performance gap is negligible across the first three categories, with decreases of less than 1% in Authors (-0.17%), Genres (-0.97%), and Materials (-0.85%), while we obtain an improvement in the Subjects category.

Strategy	Authors	Genres	Materials	Subjects
Multi-Model	**18.45**	**76.65**	**70.44**	7.00
Best Model	18.28	75.68	69.59	**8.14**
Difference	*-0.17*	*-0.97*	*-0.85*	***+1.14***

Table 6. Benchmarking the Best Configuration (1 IMG–4 PROMPT) against the Multi-Model Upper Bound (Exact Match %). The 1 IMG–4 PROMPT approach remains highly competitive while offering superior efficiency.

Finally, in Table 7 we present the comparison of our pipeline using a zero-shot VLM model and the finetuned one. The fine-tuning process yields massive improvements over the pre-trained capabilities, registering a mean percentage increase of +31.65% in Exact Match accuracy and +21.61% in ROUGE-1 scores. We observe the most significant gains in structured categories such as Genres (+69.28% EM) and Materials (+47.79% EM), where the model successfully aligns with the specific taxonomy of the dataset.

6. Conclusion

This study presents ARTLM, a multimodal framework designed for the automated generation of structured metadata from digitized artworks. The proposed pipeline formulates metadata extraction as a conditional multimodal generation task, in which a high-resolution artwork image and a category-specific textual prompt are processed by a generative multimodal model to produce semantically consistent metadata predictions.

Experimental results demonstrate that Qwen2-VL constitutes a highly effective backbone for this task, consistently outperforming the alternative multimodal models considered in our evaluation. Furthermore, fine-tuning the model through Low-Rank Adaptation (LoRA) yields a substantial improvement over zero-shot inference, with particularly notable gains in Exact Match accuracy, which serves as our primary evaluation metric. These improvements are especially evident for complex and visually grounded attributes such as Genres and Materials, highlighting the benefit of task-specific adaptation in multimodal settings.

Despite these promising results, several challenges remain. Performance degradation is observed for infrequent labels, most notably within the highly sparse Subjects category, indicating limitations in modeling long-tail distributions. In addition, the reliance on multi-prompt inference strategies, while beneficial for category-specific specialization, introduces non-negligible computational overhead at inference time. Addressing label sparsity and improving inference efficiency represent important directions for future work.

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Model	Exact Match Aut	Gen	Mat	Sub	Semantic Similarity Aut	Gen	Mat	Sub	BLEU Aut	Gen	Mat	Sub	ROUGE-1 Aut	Gen	Mat	Sub
Zero-Shot (Qwen)	15.00	6.40	21.80	1.40	80.00	58.60	88.60	**71.70**	-	7.90	21.00	3.00	15.40	32.40	60.90	19.30
Best Fine-Tuned	**18.28**	**75.68**	**69.59**	**8.14**	**86.31**	**75.68**	**96.43**	60.34	-	**20.07**	**44.60**	**4.63**	**21.25**	**80.56**	**87.33**	**24.99**
Improv.	*+3.28*	*+69.28*	*+47.79*	*+6.26*	*+6.31*	*+17.08*	*+7.83*	*-11.36*	-	*+12.17*	*+23.60*	*+1.63*	*+6.15*	*+48.16*	*+26.43*	*+5.69*

Table 7. Comprehensive comparison between Zero-Shot baseline and Best Fine-Tuned Configuration across all metrics.

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